

PROJECT: CAUSAL MACHINE LEARNING

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Q1

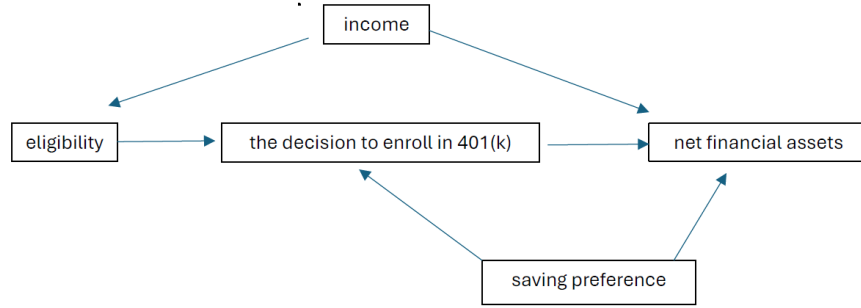


Figure 1

Q2

The ATE estimated by AIPW and TMLE is approximately \$7,800, which means eligibility for a 401(k) plan causally increases net financial assets by approximately \$7,800 on average. And AIPW and TMLE adjust for age, income, family size, years of 1 education, a married indicator, a two-earner status indicator, a defined benefit pension status indicator, an IRA (individual retirement account) participation indicator, and a home ownership indicator. These results are statistically significant and consistent across methods, lending confidence to their validity.

While the unadjusted linear model suggests that individuals eligible for 401(k) plans have on average \$19,559 more in net financial assets, with the confidence interval [16999.90 , 22118.79]. However, this estimate is likely biased upward due to confounding .

AIPW used causaldrf package, and TMLE used tmle package. All of these models used 5-fold cross-fitting.

The results of AIPW and TMLE are significantly lower than the naive unadjusted estimate, highlighting the importance of controlling for confounders in causal analysis.

Method	ATE Estimate	95% CI	p-value
Linear Regression(unadjusted)	19,559	[16,999 , 22,119]	<2e-16
AIPW	7,754	[5,098, 10,411]	< 0.001
TMLE	7,825	[5,198, 10,453]	< 0.001

Table 1 ATE estimated from different models

Q3

Expand covariance and transform it according to the law of iterated expectations:

$$\theta = E\{Cov(A, Y|L)\} = E[(A - E[A|L])(Y - E[Y|L]) | L] = E[(A - E[A|L])(Y - E[Y|L])]$$

Efficient influence curve: $IC = (A - E[A|L]) \cdot (Y - E[Y|L]) - \theta$

Procedure of calculation:

(1).Estimate $E[A|L] = g_1(L)$ and $E[Y|L] = P(L)$ using Superlearners with 5-fold cross-fitting.

(2).Compute $\hat{\theta}$ by making sample mean of influence curve equal to zero: $\hat{\theta} = \frac{1}{N} \sum_{i=1}^N (A_i - g_1(L_i))(Y_i - P(L_i))$

(3).Estimate the standard error using the sample standard deviation of the influence function: $SE(\hat{\theta}) = \frac{1}{\sqrt{N}} SD(IC_i)$

We get: $\hat{\theta} = 1699.57$; $SE(\hat{\theta}) = 264.63$; z-statistic = 6.42 and $p = 1.34 \times 10^{-10}$.

Since the p-value is extremely small ($< 10^{-9}$), we strongly reject the null hypothesis $H_0: \theta = 0$. 95% Confidence Interval is [1180.89, 2218.25]. That doesn't include 0, which indicates that the conditional covariance is significantly not equal to 0. This provides strong evidence that eligibility for a 401(k) plan is conditionally associated with net financial assets, even after controlling for covariates L.

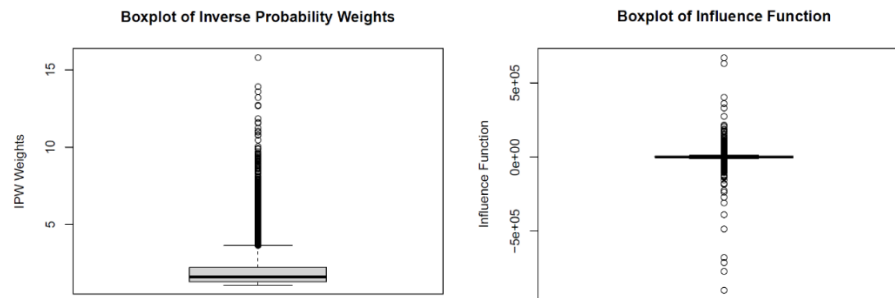


Figure 2 Boxplot of IPW and Influence function

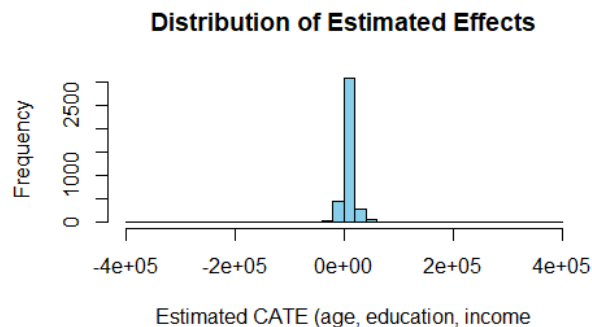
Most IPW are concentrated between 1 and 3. There are a few extreme values(>10), but the overall part is still acceptable. The values of the influence function are mostly concentrated around 0, but there are also obvious extreme values on both sides, which may affect the robustness of the results.

Q4:

We use the R-learner framework to estimate the impact of 401(k) eligibility on net financial assets, based on age, income, and education. The analysis controls for confounders from question 2. We applied 5-fold cross-fitting and the SuperLearner ensemble to estimate nuisance functions and pseudo-outcomes, with R-learner estimates obtained via weighted regression on these pseudo-outcomes.

Histogram of Estimated CATEs

This histogram is tightly centered around zero, with a majority of effects clustered near zero. This suggests that for most individuals, eligibility for a 401(k) plan has a modest effect on net financial assets, though some outliers show substantially higher or lower estimated impacts. The mean is **\$8052.422** with 95% CI = [**\$7765.665** , **\$8339.179**]



Heterogeneity by Covariates

We estimated separate CATE models for age, income, and education to visualize how treatment effects vary by individual characteristics. Each variable shows a different effect distribution due to their varying distributions.

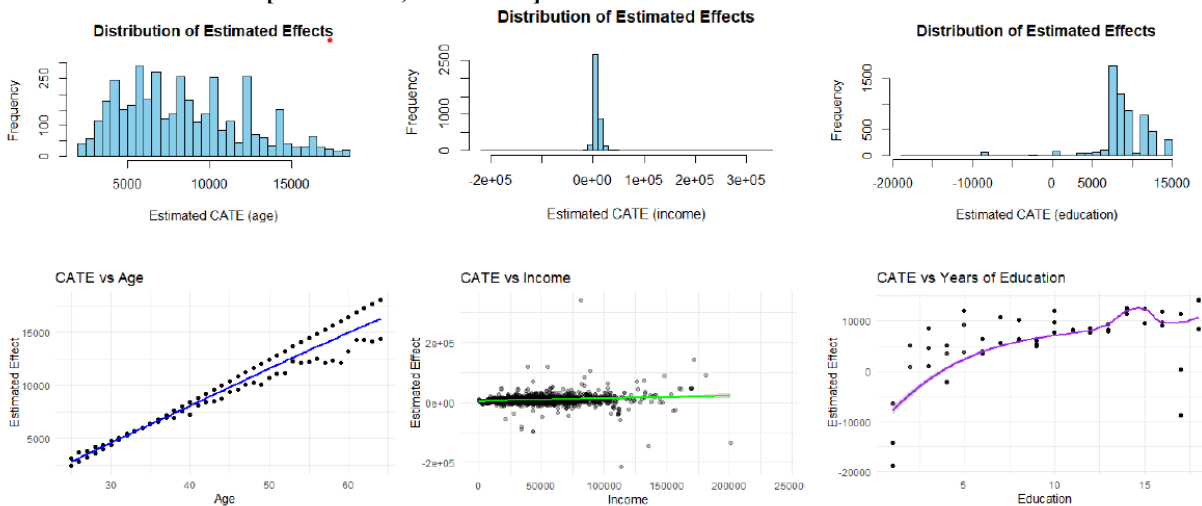
Age : the histogram shows a wide, slightly right-skewed distribution, indicating substantial heterogeneity. The scatterplot reveals that treatment effects tend to increase with age, particularly among middle-aged and older individuals, suggesting that age influences the effect of 401(k) eligibility, with older individuals benefiting more on average.

The mean is **\$6901.652** and 95% CI is [**\$6854.253** , **\$6949.051**]

Income: The distribution is centered and very narrow, indicating little variation in treatment effects across income levels. This is further supported by the flat trend in the scatterplot of CATE vs. income. The mean is **\$7008.736** and 95% CI is [**\$6961.62** , **\$7055.852**]

Education: The histogram shows moderate dispersion and right skewed, which shows that education have a big effect on treatment. The scatterplot indicates that treatment effects increase slightly with higher levels of education, though the relationship is weaker than with age. And they also have many outliers, which are some point not follow the trend. In conclusion, These findings suggest that 401(k) benefits are not evenly distributed. Older individuals and those with higher education tend to gain more, likely due to better awareness and a longer investment horizon. Income plays a smaller role, suggesting that financial knowledge and planning matter more than income alone. The mean is

\$6876.856 and 95% Ci is [\$8593.089 , \$8711.472]

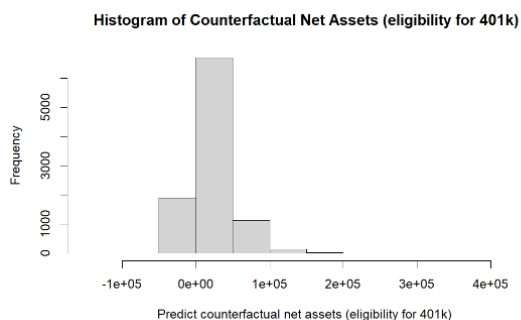


Q5:

We constructed pseudo-outcomes combining outcome modeling and inverse propensity weighting and used them to predict counterfactual outcomes ($Y1_{\tilde{d}}$). Finally, we evaluated prediction errors and constructed a 95% confidence interval using the efficient influence function (EIF). The function of EIF is

$$\frac{A}{P[A = 1|L]} \{Z - E[Z|A = 1, L]\} + E[Z|A = 1, L] - E[Z^1]$$

with $Z^1 = (Y^1 - \tilde{Y}^1)^2$ and $Z = (Y - \tilde{Y}^1)^2$

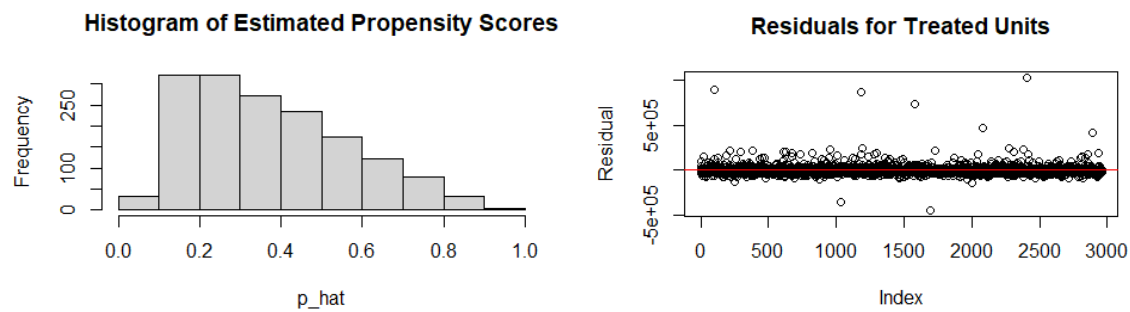


The histogram of predicted counterfactual net assets ($\tilde{Y}^1$) shows a right-skewed distribution, typical for financial data, with most individuals having low to moderate assets and a smaller group having higher net wealth. This reflects the usual wealth distribution, where a few accumulate much more, often due to higher income or longer saving histories. The large range of the x-axis (from -10^5 to 10^5) is due to the sensitivity of pseudo-outcomes in the DR-Learner methodology to treatment assignment and nuisance parameters.

The estimated prediction error for counterfactual net financial assets under 401(k) eligibility is **394923486** with a 95% confidence interval of **[-18701454794 , 19491301766]** and standard error is **9743050143**. This large error and wide interval is due to some of the reasons. First, it is because the outcome that we deal with is the money, which is often have a high value and varies a lot. Furthermore, the metric we calculate is mean square, which is can be hugely affected by some large error or outliers. The DR learner itself also is not stable and very sensitive with outlier because of the extreme pseudo-outcomes, and high variance from inverse propensity weights. Hence, the range of outcome from this model is often larger than R-learner. Some diagnostics following will strengthen the fact that DR learn is rationally give a wide confidence score and sensitive to data irregularities and should be used cautiously for predicting financial outcomes.

1. Extreme Pseudo-Outcomes:

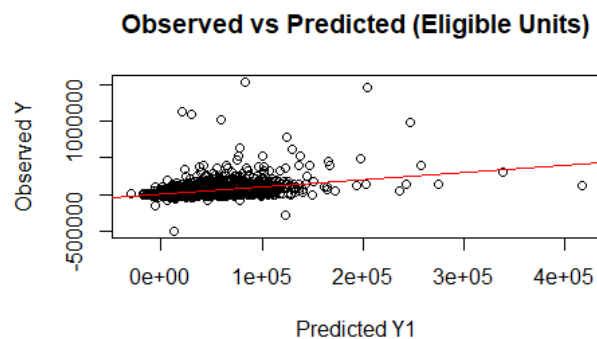
The pseudo-outcome formula can produce large values if the propensity score ($p^{\wedge}$) is very small or close to zero. The histogram shows most scores are between 0.2 and 0.8, but some are near the boundaries (min: 0.07), inflating inverse probability weights. This contributes to instability in pseudo-outcomes, with diagnostics revealing several extreme values that may cause high variance in the DR learner.



2. Influence Function:

From EIF there are 15 influential observations, which is exceeding 3 standard deviations from the mean, confirming potential outlier influence.

3. Observed vs Predicted (Eligible Units):



The comparison of predicted and observed outcomes reveals significant deviations, particularly for treated individuals. Many individuals have observed values up to 10 times higher than predicted, and surprisingly, some observed values are extremely negative while predicted values are high. These discrepancies highlight poor model fit in certain areas, leading to exceptionally large errors.

In conclusion, the DR-Learner combined with Super Learner is a theoretically strong approach, but it can be highly sensitive to outliers and extreme weights. In our case, using raw monetary values with large variance likely inflated the squared errors, resulting in high prediction error and instability. This could be mitigated

by transforming or minorizing the outcome, trimming extreme propensity scores, or using more robust models within the Super Learner. Overall, while the method is statistically sound, applying it to real-world financial data requires extra care to handle scale and outliers for more reliable results.